CAT Arithmetic Topic Tracker & Practice Strategy

8-Topic Sequenced Learning Path, Formula-Free Heuristics, Weightage Breakdown & Accuracy Targets

1. CAT Arithmetic Topic Architecture & Question Weightage

Arithmetic accounts for 35% to 45% (8-10 questions) of the entire CAT Quantitative section.

Topic Name	Historical CAT Weightage	Difficulty Spectrum	Primary Heuristic / Shortcut
Percentages, Profit & Loss	2 - 3 Questions	Easy to Moderate	Fraction-to-percentage conversions; multiplier method (1.15x)
Simple & Compound Interest	1 - 2 Questions	Moderate	Successive percentage changes; effective rate approximations
Ratios, Proportions & Variations	1 - 2 Questions	Easy to Moderate	Constant of proportionality; multiplying factor heuristics
Averages, Mixtures & Alligations	2 - 3 Questions	Moderate to High	Teeter-totter balance method; deviation from assumed mean
Time, Speed and Distance (TSD)	2 - 3 Questions	Moderate to High	Relative speed, circular tracks, trains, escalators & boats
Time and Work (Pipes & Cisterns)	1 - 2 Questions	Easy to Moderate	LCM efficiency method; fractional rate of completion
Installments & Commercial Math	0 - 1 Question	Moderate to High	Geometric progression formula for equal annual installments

STRATEGIC PRINCIPLE: Milestone: Mastering Arithmetic alone guarantees clearing the 85-90th percentile sectional cutoff in QA.

2. The 4-Stage Arithmetic Practice Protocol

Progress from conceptual understanding to high-speed exam-level execution.

Learning Stage	Question Source / Material	Target Accuracy	Time Limit per Question
Stage 1: Concept Drills	Foundational textbook problem sets (30 Qs/topic)	90%+	Untimed (Focus on concept understanding)
Stage 2: Formula-Free Solving	Past 10 years actual CAT Arithmetic questions	80%+	2.5 Minutes (Eliminate algebraic equations)
Stage 3: Mixed Topic Sprints	20-question randomized Arithmetic tests	75%+	2.0 Minutes (Test-day cognitive switching)
Stage 4: Elimination & Option Testing	Difficult Level-3 Arithmetic trap questions	85%+	1.5 Minutes (Back-solving with options / substitution)

STRATEGIC PRINCIPLE: Key Technique: Always replace x and y with realistic numbers (100, 10, LCM) whenever possible.

3. Arithmetic Error Taxonomy & Weekly Review Tracker

Track and eradicate the three primary sources of lost Arithmetic marks.

- Error 1: Percentage Base Confusion. Forgetting whether percentage increase is based on cost price or selling price.
- Error 2: Units Discrepancy in TSD. Mixing km/hr with m/s or minutes with hours in multi-stage speed calculations.
- Error 3: Sign Errors in Alligation Diagrams. Inverting mean deviations when calculating replacement mixture ratios.